

1.3 Complex Numbers

In algebra, we have worked primarily with real numbers. However, certain equations, such as $x^2 = -1$, have no solution within the real numbers. To handle such cases, mathematicians defined a new type of number involving the square root of negative values.

Definition

A **complex number** is a number written in the form $a + bi$, where a and b are real numbers and i is the imaginary unit defined by

$$i = \sqrt{-1}, \quad i^2 = -1.$$

If $b = 0$, the number is real; if $a = 0$, it is purely imaginary. The value a is called the **real part**, and b is the **imaginary part**.

The imaginary unit allows us to express square roots of negative numbers. For instance:

$$\sqrt{-b} = i\sqrt{b}, \quad \text{for } b > 0.$$

This leads to a consistent extension of the real number system, forming what we call the **complex number system**.

Example: Writing expressions in terms of i . Write each expression in terms of i .

$$\text{a) } \sqrt{-25} \quad \text{b) } \sqrt{-12} \quad \text{c) } \sqrt{-13}$$

Solution:

$$\text{a) } \sqrt{-25} = i\sqrt{25} = 5i.$$

$$\text{b) } \sqrt{-12} = i\sqrt{12} = i\sqrt{4 \cdot 3} = 2i\sqrt{3}.$$

$$\text{c) } \sqrt{-13} = i\sqrt{13}.$$

Example: Writing complex numbers in standard form. Simplify each expression and write in the form $a + bi$.

$$\text{a) } 3 - \sqrt{-100} \quad \text{b) } \frac{2 + 7i}{5} \quad \text{c) } \frac{-6 + \sqrt{-18}}{9}$$

Solution:

$$\text{a) } 3 - \sqrt{-100} = 3 - 10i = 3 + (-10i).$$

$$\text{b) } \frac{2 + 7i}{5} = \frac{2}{5} + \frac{7}{5}i.$$

$$\text{c) } \frac{-6 + \sqrt{-18}}{9} = \frac{-6 + 3i\sqrt{2}}{9} = -\frac{2}{3} + \frac{\sqrt{2}}{3}i.$$

Definition

Powers of i : The powers of i repeat every four terms:

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1.$$

The pattern repeats as $i, -1, -i, 1, i, -1, -i, 1, \dots$

Example: Simplifying powers of i . Simplify each expression.

$$\text{a) } i^{48} \quad \text{b) } i^{23} \quad \text{c) } i^{50} \quad \text{d) } i^{-19}$$

Solution:

$$\text{a) } i^{48} = (i^4)^{12} = 1^{12} = 1.$$

$$\text{b) } i^{23} = i^{20} \cdot i^3 = (1) \cdot (-i) = -i.$$

$$\text{c) } i^{50} = i^{48} \cdot i^2 = (1)(-1) = -1.$$

$$\text{d) } i^{-19} = i^{-20} \cdot i = (i^4)^{-5} \cdot i = (1) \cdot i = i.$$

Real-World Example: AC Circuits

In electrical engineering, alternating current (AC) is modeled using complex numbers. The voltage across a circuit may be expressed as $V = 120 + 30i$ volts, where the real part represents the resistive (in-phase) component, and the imaginary part represents the reactive (out-of-phase) component.

Solution: The magnitude (or RMS voltage) is given by:

$$|V| = \sqrt{(120)^2 + (30)^2} = \sqrt{15300} = 123.7 \text{ volts.}$$

Extra Practice

Simplify or write in standard form.

$$1) \sqrt{-45}$$

$$2) 4 - \sqrt{-9}$$

$$3) \frac{-10 + \sqrt{-16}}{4}$$

$$4) i^{34}$$

$$5) i^{-7}$$

$$6) 2i^5 - 3i^2 + i$$